

Lecture 1

Overview, Labor Demand and Supply

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40073 Monetary Economics and the Macroeconomy
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General Presentation

- **Who am I?** I am a macroeconomist working on
 - Optimal fiscal policies in presence of income and wealth inequality
 - Labor market dynamics and government policies: minimum wage, parental leave
- Some of my published and working papers:
 - *Taxation of Top Incomes and Tax Avoidance* (International Economic Review 2025)
 - *Rescue Policies for Small Businesses during the COVID-19 Recession* (Review of Economic Dynamics, 2023)
 - *Gendered Effects of the Minimum Wage* (International Economic Review 2026)
- **Who are you?** Econ background?

Course Information

- Lecturer: Alessandro Di Nola
- Office: University House, 2nd floor - 2137
- E-mail: a.dinola@bham.ac.uk
- Office hours: via e-mail
- Discussion Board: for questions including assessment, you can write on the Discussion Board on Canvas.
- Course schedule:
 - Lectures on Tuesday and Thursday
 - Seminar classes (check timetable)
- Assessment:
 - Written exam (50%)
 - Problem Set (25%): requires coding in MATLAB or Python
 - Essay (25%)

This Course: Modern Macroeconomics

- Modern = as practiced in current academic research and policy analysis
 - $\neq$ what's in standard first-year textbooks
- Macro with micro foundations
 - Utility and profit maximization just like in microeconomics
 - But general equilibrium $\implies$ prices endogenous, clear markets
- Core topics in macro: general equilibrium models, from simple one period model to real-business cycle model, fiscal policy, monetary frictions, new-Keynesian model

Topics we will cover (subject to change)

- **Section 1:** Welcome and Overview, Labor Demand and Supply
- **Section 2:** A Simple Macro Model, Equilibrium and Welfare Theorems
- **Section 3:** Consumption, Saving, Interest Rates
- **Section 4:** Investment and Capital Accumulation
- **Section 5:** Business Cycle Macro and Lucas Critique
- **Section 6:** New Keynesian Model I
- **Section 7:** New Keynesian Model II
- **Section 8:** The Financial Crisis, Asset Bubbles
- **Section 9:** Unemployment (Pissarides), Inequality in Macro

Tools

As just mentioned: macro with **micro foundations**

- utility and profit maximization familiar from microeconomics

Reflected in tools:

- differential calculus
- heavy use of **constrained optimization using Lagrangean methods**
- Need to know how to solve

$$\max_{c_1, c_2} u(c_1, c_2) \quad \text{s.t.} \quad p_1 c_1 + p_2 c_2 \leq m$$

using

$$\mathcal{L} = u(c_1, c_2) + \lambda [m - p_1 c_1 - p_2 c_2]$$

- If you don't, refresh your memory

Textbooks

Unfortunately, there exists no textbook with the proper level and focus for this course. Instead, it will rely on lecture notes and research journal articles

That being said, the following textbooks may be useful as background reading:

- **Kurlat, Pablo, *A Course in Modern Macroeconomics***

`https:`

`//sites.google.com/view/pkurlat/a-course-in-modern-macroeconomics`

- will also post relevant PDF excerpts on Canvas
- Williamson, Stephen, *Macroeconomics*.

How to Study Efficiently

- The slides are the most important. Then required readings.
- Keep up with the course. Much of the material is new and needs time to sink in, and material taught later builds on material covered earlier.
- Spending some time every week will help you
 - absorb and understand the material in fewer hours than catching up later,
 - enjoy the lectures more because you understand more, and
 - reduce stress.

How to Study Efficiently

- Make **good notes**. Revision before the exam will be easier.
- Come to class prepared, having studied the course material and the assignment.
- Be an **active learner**:
 - Do not just follow the material line by line. Put it away and see whether you can rebuild the argument from scratch.
 - Explain what you have learned to someone else, such as a friend, fellow student, etc.
 - Study with other students and challenge each other.
 - When you get lost, do not immediately search for more literature. The struggle to work things out yourself is part of studying effectively.

Some thoughts on methodology:
what economists do and why

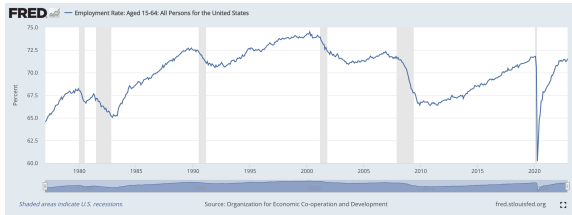
The interplay of models and data

- Like in your previous courses, will make use of models and look at data
- Example of model: firms choose labor demand to maximize profits

$$\max_n f(n) - wn$$

where n is labor and $f(n)$ is output (f is the production function). The consumption good's price is normalized to one, so w is the real wage.

- Example of data:



The interplay of models and data

How to think about interplay of model and data?

- why not only data?
- why not only models?

Here: quick, superficial treatment of important methodological issues

- part of much broader methodological debates (see references)
- my main goal here: get you thinking about these issues
- disclaimer: no single “right” approach here, reflects my personal opinion

The interplay of models and data

Macroeconomics needs interplay of both theory and data

1. Theory without data

- “smart kids playing in sandbox”

2. Data without theory

- flying blind
- “Let the data speak” is fiction / wishful thinking
- there is no paper in economics without a (implicit) model
- policy questions are questions about counterfactuals (“what happens in world with policy relative to world without policy?”)
- ... requires a model (either explicit or implicit)

Further reading for the interested (not required)

Some relevant references (discussion in many much deeper than here):

- Nakamura-Steinsson “Identification in Macroeconomics”
- Reis “Is something really wrong with macroeconomics?”
- JEP Symposium “Con out of Economics”
- Card “Model-Based or Design-Based? Competing Approaches in ‘Empirical Micro’”

<https://www.aeaweb.org/issues/126>

<https://davidcard.berkeley.edu/lectures/woytinsky.pdf>

<https://www.youtube.com/watch?v=S6xSEiB6E2s>

Modeling in (Macro)economics

Modeling in (Macro)economics

Objective is **not** to build one big model we use to address all issues

- descriptive realism is not the objective
- instead make modeling choices that are dependent on the issue
- whether a model is “good” is context dependent

Approach to modeling in (macro)economics well summarized by

- “All models are false; some are useful”
https://en.wikipedia.org/wiki/All_models_are_wrong
- “If you want a model of the real world, look out the window” (kidding, but only half kidding)
- “The map is not the territory”
https://en.wikipedia.org/wiki/Map-territory_relation
- Literary version of this: Borges “On Exactitude in Science”
- Point in case: geographically accurate London tube map
<https://kottke.org/plus/misc/images/tubegeo.gif>
<https://www.ifs.org.uk/publications/10282> by Bandiera, Machin,...

Toward a Simple Macro Model: Labor Supply and Demand

Toward a simple macro model

- Next lecture: your first modern macro model
- Model will be extremely simple
 - a single “representative” consumer
 - a single “representative” firm
 - only one time period, i.e. model is static not dynamic
 - only one good
 - only one factor of production (labor)
 - no frictions, “market imperfections”
 - model is sometimes called **“Robinson Crusoe economy”**
- ... but it will have key features of most modern macro models
 1. micro foundations
 2. general equilibrium
- Rest of course: build much richer, more satisfactory models

Sketch of simple macro model (Robinson Crusoe economy)

- **Preferences** (i.e. utility function):

$$U(c, h)$$

- c : consumption of coconuts, U is increasing in c
- h : hours worked, U is decreasing in h

- **Technology** (i.e. production function):

$$y = f(n)$$

- y : output, i.e. production of coconuts
- n : hours employed, f is increasing in n

- **Resource constraints:**

$$c = y, \quad n = h$$

- Next lecture: different ways of organizing this economy
- One of them: **Robinson Crusoe** is alone on an island, lives off coconuts, harvesting coconuts takes time. Hence the name.

Some thoughts on the representative agent assumption

- Model assumes “representative consumer” and “representative firm” (jointly = “representative agent”)
- When is this justified? Here: sketch, more in supplement (not examinable)
- **Firms**: if at least one of following 3 conditions is satisfied
 1. all firms in economy are identical
 2. particular assumptions on production functions (“homogeneity”)
 3. perfect factor markets (capital, labor) $\Rightarrow$ equalize marginal products
- **Consumers** (even harder): if one of following 2 conditions is satisfied
 1. all consumers in economy are identical
 2. particular assumptions on preferences (“Gorman aggregation”)
- Do we believe these conditions are satisfied? **No**, but...

Toward a simple macro model

Today: study key building blocks of simple macro model

1. labor demand
2. labor supply

This should largely be a refresher of microeconomics

- probably with different notation
- good point to mention: you need to be able to handle changing notation
 - economists change notation all the time
 - notation varies across textbooks, courses, even within lectures
- my experience: if you can't handle changing notation, you probably don't understand the economics

Vocabulary: “partial equilibrium” and “general equilibrium”

Will frequently hear macroeconomists use following two words

- “general equilibrium” = all prices endogenously determined in equilibrium
- “partial equilibrium” = problem of individual household or firm in isolation taking as given prices

Let's apply vocabulary:

- Today: labor demand and supply in partial equilibrium
- Next lecture: labor demand and supply in general equilibrium

More vocabulary: Endogenous variables vs exogenous variables

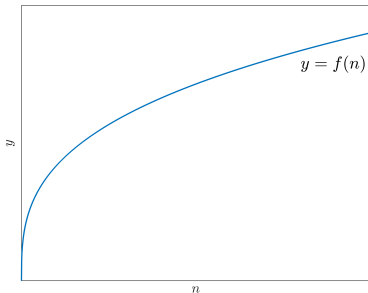
- Endogenous variables are variables determined **within the model**, as the result of agents optimizing and market clearing
- Exogenous variables are variables determined **outside of the model**
- In Robinson Crusoe economy: consumption, labor and output are endogenous. Technology is exogenous.

Labor demand

- Representative firm has production function

$$y = f(n)$$

- Assumptions: $f'(n) > 0$ and $f''(n) < 0$ for $n > 0$



- Example: $f(n) = An^\alpha$ with $0 < \alpha < 1$, $A > 0$

Labor demand

- Representative firm maximizes profits

$$\Pi = \max_n f(n) - wn$$

- Solution: labor demand function

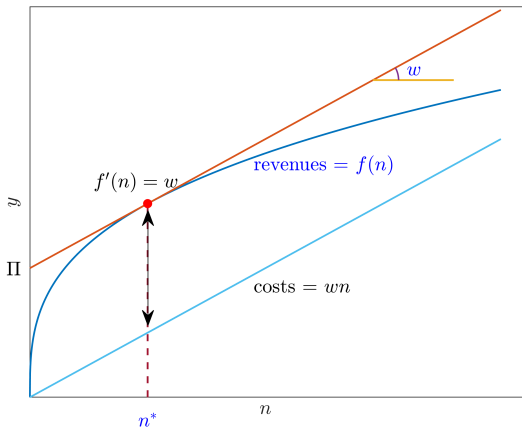
$$n^* = n^d(w)$$

- Optimality condition

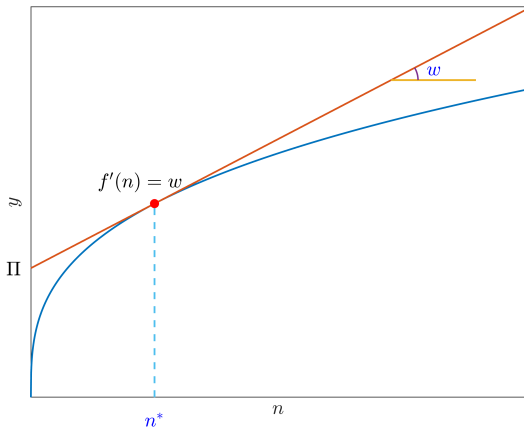
$$f'(n) = w$$

In words: Marginal product of labor equals the wage

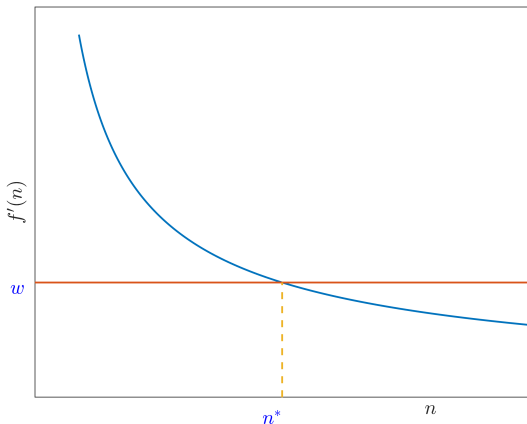
Labor demand: graphical representation



Labor demand: more minimal graphical representation



Labor demand: alternative graphical representation



- Marginal product of labor $f'(n)$ is decreasing

Labor demand: parametric example

- Example: $f(n) = An^\alpha$ with $0 < \alpha < 1, A > 0$
- Representative firm maximizes profits

$$\Pi = \max_n f(n) - wn$$

- $\Rightarrow$ optimal labor demand is

$$n^* = n^d(w) = \left(\frac{\alpha A}{w} \right)^{\frac{1}{1-\alpha}}$$

which is

- decreasing in wage w (as a demand curve typically is)
- increasing in productivity A

Labor Supply: Preferences

- Preferences

$$u(c) - v(h)$$

where c = consumption, h = hours worked, $u(c)$ = utility from consumption, and $v(h)$ = disutility from work

- Assumptions

- $u'(c) > 0$ and $u''(c) < 0$
 - $v'(h) > 0$ and $v''(h) > 0$, i.e. disutility from work is increasing and convex
- Example 1: $u(c) = \log c$, $v(h) = -\theta \log(1 - h)$, $\theta > 0$
 - Example 2:

$$u(c) = \log c, \quad v(h) = \theta \frac{h^{1+1/\varepsilon}}{1 + 1/\varepsilon}, \quad \theta, \varepsilon > 0$$

Labor Supply: Household Problem

Representative household solves

$$\max_{c,h} u(c) - v(h) \quad \text{s.t.} \quad c = wh + \Pi$$

Explanation:

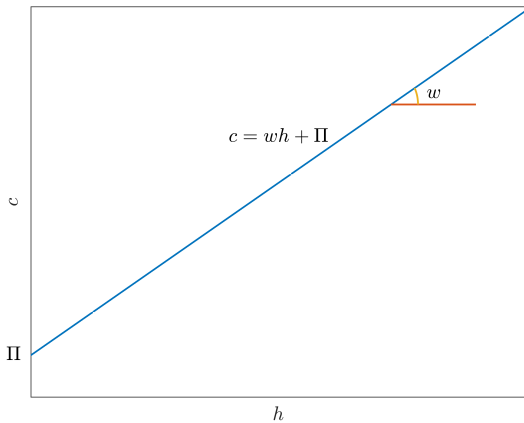
- c consumption, h hours worked
- w = real wage per hour, measured in units of the consumption good
- wh = labor income
- Π = other unearned income, e.g. dividend income from owning stocks
- total available time is normalized to 1, so leisure is $\ell = 1 - h$

Solution: labor supply function

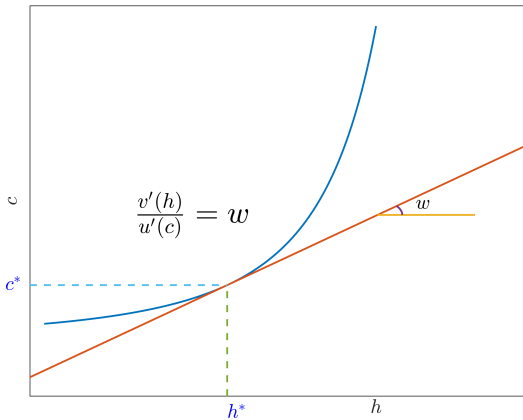
$$h^* = h^s(w)$$

Nonlabor income Π is held fixed when deriving labor supply as a function of w .

Labor Supply: Budget Constraint



Labor Supply: Optimal Choice



Derivation of optimality condition

You should already know this but I will do derivation once

Two equivalent strategies:

1. Direct substitution (brute force)

$$\max_h u(\underbrace{wh + \Pi}_c) - v(h) \Rightarrow u'(c)w - v'(h) = 0 \Rightarrow u'(c)w = v'(h)$$

2. Lagrangean

$$\mathcal{L} = u(c) - v(h) + \lambda [wh + \Pi - c]$$

$$\left. \begin{array}{l} c : u'(c) = \lambda \\ h : v'(h) = \lambda w \end{array} \right\} \Rightarrow u'(c)w = v'(h)$$

Optimal Labor Choice: Intuition

- The first-order condition is

$$u'(c)w = v'(h).$$

- But do you understand what this equation is saying?
- It is useful to adopt a **variational** perspective to build intuition.
 - Suppose the household is considering working h hours.
 - To assess this choice, suppose it contemplates working a little bit more (or a little bit less).

Optimal Labor Choice: Intuition

$$u'(c)w = v'(h)$$

- Marginal cost of working a little bit more:
 - $v'(h)$ “utils” per extra hour worked.
- Marginal benefit of working a little bit more:
 - w coconuts per extra hour worked,
 - each coconut is worth $u'(c)$ “utils,”
 - so the overall marginal benefit is $u'(c)w$ “utils” per extra hour worked.
- $u'(c)w = v'(h)$ says that, at the optimum, the marginal benefit of working more equals the marginal cost of working more.
- If either $u'(c)w > v'(h)$ or $u'(c)w < v'(h)$, then the labor choice is not optimal! (See next slide)

Optimal Labor Choice: Intuition

- Suppose the household is supplying very few hours of labor.

- What does this imply about u' ?
- What does this imply about v' ?
- In this situation, we have:

$$u'(c)w > v'(h)$$

- The worker is better off if they work more.

- Suppose the household is supplying a very large number of hours.

- What does this imply about u' ?
- What does this imply about v' ?
- In this situation, we have:

$$u'(c)w < v'(h)$$

- The worker is better off if they work less.

An exercise and a question

- Exercise: derive optimal labor supply in example 2

$$u(c) = \log c, \quad v(h) = \theta \frac{h^{1+1/\varepsilon}}{1+1/\varepsilon}, \quad \theta, \varepsilon > 0$$

- **Question:** suppose w increases, what will happen to labor supply? Why?

A way of thinking about income and substitution effects

- Recall: representative household solves

$$\max_{c,h} u(c) - v(h) \quad \text{s.t.} \quad c = wh + \Pi$$

- Optimal $h^*(w)$ solves optimality condition and budget constraint

$$\frac{v'(h)}{u'(c)} = w$$
$$c = wh + \Pi$$

- Suppose wage increases. What will happen to $h^*(w)$?

Income and Substitution Effects

1. Substitution effect:

- Working is more lucrative.
- Leisure is more “expensive.”
- The wage is the opportunity cost of leisure.
- Households substitute towards lucrative activities and away from expensive activities.
- Work more.

2. Income effect:

- A higher wage means the worker is richer.
- Richer workers choose more consumption and more leisure.
- Leisure is a “normal” good (you buy more of it the richer you are).
- More leisure means less work.
- Work less.

Income and Substitution Effects

$$\frac{v'(h)}{u'(c)} = w$$

- To isolate the incentive to work more, consider an increase in the wage holding consumption fixed.
- An increase in the wage increases the RHS.
- We need to increase the LHS to balance.
- If c is constant, $v'(h)$ must rise.
- This implies that h must rise since $v''(h) > 0$.

Income and Substitution Effects

$$\frac{v'(h)}{u'(c)} = w$$

- How does an increase in consumption affect labor supply **holding wages fixed**? (E.g., due to winning the lottery.)

$$\uparrow c \Rightarrow \downarrow u'(c) \Rightarrow \uparrow \frac{v'(h)}{u'(c)} \quad (u''(c) < 0)$$

- To balance this effect, we must have

$$\downarrow v'(h) \Rightarrow \downarrow h \quad (v''(h) > 0)$$

- Thus, the income effect leads to a fall in hours worked.

Income and Substitution Effects

- Recall that $c = wh + \Pi$ (in our simple model).

- So, we have:

$$\frac{v'(h)}{u'(wh + \Pi)} = w$$

- This means that an increase in the wage has both an income and a substitution effect.
 - w on the RHS is responsible for the substitution effect.
 - w on the LHS (inside $u'(wh + \Pi)$) is responsible for the income effect.
- Which is stronger?

Labor Supply in the Data

Real Wages and Hours Worked per Worker

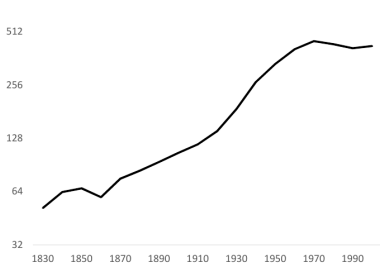


Figure 2: Real Wages in the United States

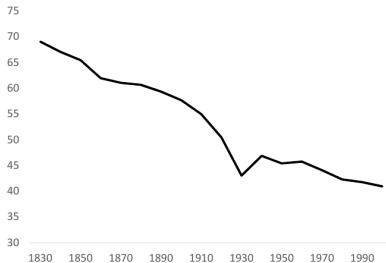


Figure 3: Weekly Hours Worked per Worker in the United States

- Real wages have risen by roughly 1.5% per year for over 100 years
- Hours worked per worker in the U.S. have declined gradually

Labor supply in practice?

- What do these figures imply about relative size of income and substitutions effects in the long run?
 - Income effect slightly larger than substitution
- Common to ignore income effects and assume wage increases will increase labor supply (e.g., tax cuts)
- But income effects are large over the long run!!

Hours worked per person

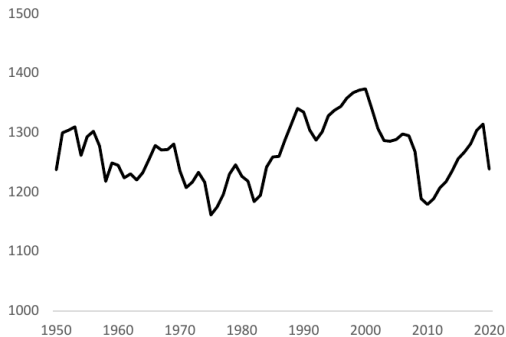


Figure 4: Annual Hours Worked per Person in the United States

- We have seen that hours worked per worker have fallen
- But hours worked per person have not
- Why might this be?

Hours worked per person

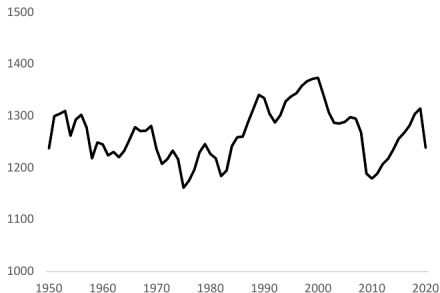


Figure 4: Annual Hours Worked per Person in the United States

- Note that

$$\frac{\text{Total hours worked}}{\text{Population}} = \frac{\text{Total hours worked}}{\text{Workers}} \times \frac{\text{Workers}}{\text{Population}}$$

Gender Revolution

- The labor force participation rate of men has fallen steadily.
- The labor force participation rate of women rose rapidly in the second half of the twentieth century.
- Changing cultural norms and less discrimination led women to enter the labor force.
- Rising female employment helped offset declining hours per worker.

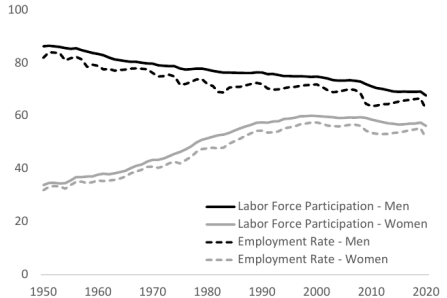


Figure 5: Employment of Men and Women in the United States

Europe vs. U.S.: Annual Hours per Working-Age Person

- In the 1950s and 1960s, annual hours per working-age person were higher in France, Germany and Italy than in the U.S.
- Hours per working-age person fell dramatically in these countries.
- Since 1980, hours per working-age person have been considerably lower in these countries than in the U.S.

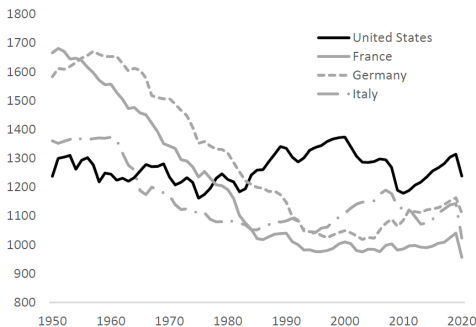


Figure 12: Annual Hours Worked per Working-Age Person

Employment Rates by Age Relative to the U.S.

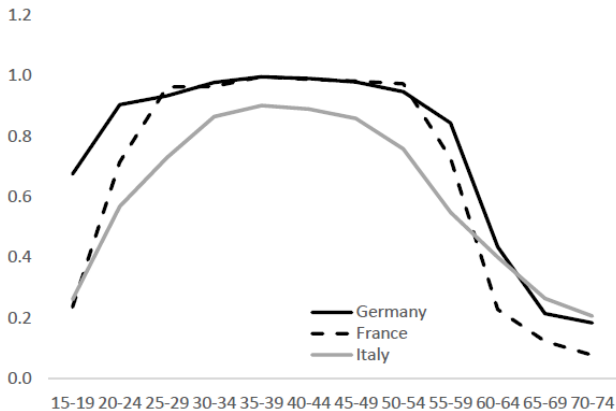


Figure 13: Employment Rates by Age Relative to the U.S.

A value of 1 means the same employment rate as in the U.S. for that age group.

Next time

- Combine labor supply and labor demand
- $\Rightarrow$ your first (simple) modern macro model